



LSH

Local Smart Home — Unified Automation Dashboard

SELF-HOSTED · OPEN PLATFORM

Works fully offline · Zero cloud dependency · Apple HomeKit native

One dashboard. Every device. No cloud required.

LSH is a self-hosted Node.js platform that unifies Victron Energy, KNX bus, IP cameras, HVAC, lighting, alarms, and 20+ smart home ecosystems into a single, real-time web interface — with native Apple HomeKit bridge, on-demand RTSP streaming, and full REST API included.



100 % local by default

All data stays inside your LAN. MQTT and Modbus run directly — no relay server, no account required. Optional VRM cloud fallback for remote access when you need it.



Protocol-agnostic architecture

MQTT, KNXnet/IP, Modbus TCP, REST, WebSocket, RTSP — add any device using standard protocols. New integrations are pluggable Node.js modules.



Native Apple HomeKit

Every sensor, switch, camera, thermostat, and relay is automatically exposed to the Apple Home app. Control your entire home from Siri or your iPhone.

ARCHITECTURE AT A GLANCE



VictronMQTT / VRM



KNX busKNXnet/IP



ESPHomeNative API



Zigbee/Z-WaveSmartThings



IP CamerasRTSP / UniFi



HVAC / LGThinQ cloud



LSH Server Node.js · Socket.IO · MQTT · REST API



Web UIAny browser



Apple HomeKitHAP bridge



REST APIAutomations



RTSP proxyLoxone / VLC



SIP softphoneVoIP browser



Victron Energy Hub

Deep integration with Victron CCGX/Venus OS via local MQTT. Monitor solar MPPT chargers, battery bank SoC, grid import/export, AC/DC loads, and inverter state in real time. Relay outputs controlled directly from the dashboard — trigger based on SoC thresholds or manual override.

- Live kW/kWh metrics with trend arrows
- MQTT → VRM cloud automatic failover
- SolarEdge production data overlay



Camera & Video

UniFi Protect cameras are automatically discovered and displayed as live thumbnails. Any RTSP camera (Hikvision, Dahua, Reolink...) can be re-streamed on-demand via the built-in FFmpeg RTSP proxy — each camera gets its own port, stream starts only when a client connects.

- On-demand FFmpeg re-streaming (no idle bandwidth)
- Loxone Intercom & camera app compatible
- UniFi Protect motion event push



Rooms & Sensors

Group sensors, lights, and switches into rooms with a drag-and-drop editor. Temperature, humidity, CO₂, motion, occupancy, door/window contacts, and air quality all appear in a unified timeline. Room cards update in real time via Socket.IO — no page refresh needed.

- Room modal editor — assign any sensor to any room
- Live value badges with color-coded thresholds

- Historical mini-charts for temperature / humidity



Lighting & Switches

Control IKEA, Shelly, Zigbee, KNX, ESPHome, and BoneIO lights and switches from a single panel. Dimmers, color temperature, RGB, scenes — all surfaced as HomeKit accessories automatically. Tactile feedback and instant state sync via MQTT or native API.

- Brightness + color temperature sliders
- SmartThings, Dirigera & Trådfri supported
- BroadLink IR/RF scene macros



MQTT Explorer

Built-in live MQTT topic browser with subscribe and publish panels. Inspect any topic value as it updates, send test commands, and onboard custom DIY devices without leaving the dashboard. Separate connection from the Victron bus — safe to use simultaneously.

- Topic tree with live value display
- Retained message inspection
- SmartBob & DIY MQTT device support



Security & Access

Built-in user management with bcrypt password hashing. Automatic Let's Encrypt TLS with hot certificate renewal — no restarts. Satel Integra alarm panel integration for zone status, arm/disarm, and sensor display. Optional HTTP → HTTPS redirect on the standard port.

- Let's Encrypt auto-renew with ACME HTTP-01
- Manual cert / self-signed TLS supported
- Satel Integra alarm zones & partitions

20+

Integrations

15+

IoT protocols



HomeKit accessories

0

Cloud dependencies

< 60s

Setup to live data

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Local Smart Home Platform

Node.js 18+ · Raspberry Pi 4 / Linux x64
No subscription · No cloud lock-in · Page 1 of 3

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Integration Reference — All supported devices & protocols

20 + INTEGRATIONS

Energy · Security · Lighting ·
Climate · Cameras · Buses

https://lsh.home:3001 — Live Dashboard

SOLAR MPPT

3.4 kW

↑ generating · 108 Voc

BATTERY SOC

87 %

Charging · 42.1 V · +38 A

GRID

-0.8 kW

Exporting surplus

AC LOAD

1.2 kW

5 circuits active

RELAY CONTROL

Heat pump



EV Charger (11 kW)



Garden irrigation



LIVING ROOM

21.4°C

68 % RH · CO₂ 612 ppm

BEDROOM

19.8°C

55 % RH · 🌙 occupancy

CAMERAS — 3 ONLINE · RTSP PROXY ACTIVE



Front Door



Garden



Garage

COMPLETE INTEGRATION REFERENCE

INTEGRATION	PROTOCOL	WHAT YOU GET	HOMEKIT TYPES
⚡ ENERGY & SOLAR			
Victron CCGX	MQTT	Real-time solar, battery, grid, AC/DC load data. Relay 1 & 2 control. Automatic VRM cloud fallback when local MQTT unreachable.	Temperature Switch
Victron VRM	HTTPS REST	Cloud fallback: live values polled from VRM portal when MQTT broker at 192.168.x.x is unreachable. No extra config needed.	—
SolarEdge	REST API	PV production data from SolarEdge cloud API. Site overview, inverter power, energy today/this month. Polled every 15 min.	Sensor
🏠 SMART HOME HUBS			
Samsung SmartThings	REST API	All paired devices: lights, switches, sensors, thermostats, locks, cameras, presence detection. Polled every 30 s with live push.	Switch Light Motion Thermostat
IKEA Dirigera	WebSocket	IKEA smart home hub — lights (on/off, brightness, CCT, RGB), remotes, motion sensors, air quality sensors via local WebSocket API.	Light Motion Air Quality
IKEA Trådfri	CoAP/DTLS	Legacy IKEA Trådfri gateway. Lights, remotes, motion sensors via CoAP. Requires node-tradfri-client npm package.	Light Motion
Homey	REST + WS	Athom Homey Pro — all Homey devices including Z-Wave, Zigbee, 433 MHz, and cloud bridges. Real-time updates via Homey Web API.	Switch Light Motion Sensor
Fibaro Home Center	REST API	Fibaro HC2/HC3 — scenes, devices, room assignments. Z-Wave devices polled via local LAN REST API with live value updates.	Switch Light Sensor
🔌 BUSES & CONTROLLERS			
KNX / KNXnet-IP	KNXnet/IP UDP	Full KNX bus integration over IP gateway. Configure group addresses with DPT type (DPT1 boolean, DPT5 1-byte, DPT9 2-byte float, DPT14 IEEE). Read on connect	Switch Light Sensor

INTEGRATION	PROTOCOL	WHAT YOU GET	HOMEKIT TYPES
		+ live GroupValue_Write events. Write commands back to bus.	
Loxone Miniserver	WebSocket	Loxone Miniserver local API — virtual inputs/outputs, rooms, sensors, lighting scenes. Full bidirectional control. RTSP proxy streams Loxone-compatible camera URLs.	Switch Sensor
BoneIO	MQTT	Polish I/O controller over MQTT. Digital inputs, relay outputs, analog sensors. Per-entity topic mapping — registers each I/O as an individual sensor/switch.	Switch Sensor
Waveshare Modbus	Modbus TCP	Waveshare relay board and I/O expansion modules. Reads coils and registers over Modbus TCP. Multiple device support with configurable register maps.	Switch
SmartBob	MQTT	Generic MQTT controller — configurable per-entity state and command topics. Supports switches, lights, temperature, humidity, numeric, and boolean sensor types with optional MQTT broker auth.	Switch Light Sensor
🔧 CLIMATE & APPLIANCES			
LG ThinQ	ThinQ REST	LG smart appliances — air conditioners, heat pumps, washing machines, dryers, dishwashers. Real-time status via LG ThinQ cloud API with token refresh.	Thermostat Sensor
ESPHome	Native API	ESPHome DIY devices over local native API (TCP). Sensors, switches, lights, binary sensors, climate entities auto-discovered. No MQTT broker needed.	Switch Light Sensor Thermostat
Dreame	Miio LAN	Dreame robot vacuum cleaners over local Miio protocol. Status, battery, room cleaning zones. Start/stop/dock commands via dashboard.	Sensor
🔒 SECURITY			
Satel Integra	ETHM TCP	Satel Integra alarm panel via ETHM-1 Plus Ethernet module. Zone states (triggered, tamper, masked), partition arm/disarm with code, zone names configurable.	Motion Contact Security
UniFi Protect	WebSocket	Ubiquiti UniFi Protect NVR — camera live thumbnails, motion events, smart detection (person/vehicle/package). Cameras appear as HomeKit Secure Video sources.	Camera Motion
📺 CAMERAS & AV			
FFmpeg RTSP Proxy	RTSP	On-demand FFmpeg re-streaming for any RTSP source. Each camera gets a dedicated port (base+N). Stream starts only when a client connects — zero idle bandwidth. Loxone Intercom compatible.	—
Shelly	MQTT / HTTP	Shelly Gen1/Gen2 relay modules, dimmers, and plugs. Real-time MQTT state updates and HTTP control. Power monitoring where available.	Switch Light
BroadLink	RF/IR LAN	BroadLink RM4 Pro / RM Mini — IR and 433 MHz RF scene control. Pre-learned codes sent as scenes or automation triggers from the dashboard.	Switch

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Local Smart Home Platform

All integrations auto-register to HomeKit · Settings UI — no YAML required

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Technical Reference — REST API, Config, Deployment

DEVELOPER REFERENCE

REST API · HomeKit · TLS ·
Config schema

GETTING STARTED IN 3 STEPS

1

Clone & Configure

Copy `config.example.json` → `config.json`. Fill in your MQTT broker address and/or VRM credentials. Set your HomeKit PIN. Run `node server.js` — live data appears within seconds.

2

Connect Integrations

Open `/settings.html` in your browser. Each integration has a dedicated card with a one-click connectivity test. No config file editing — fill in the form and click Save. New devices appear in the dashboard immediately.

3

Add to Apple Home

Open the HomeKit QR code from Settings. Scan with your iPhone — all connected devices instantly appear in Apple Home. Control them via Siri, automations, or the Home app. Bridge survives server restarts — no re-pairing.

MINIMAL CONFIG.JSON

```
{ "mqtt": { "host": "192.168.1.109", "portalId": "abc123" }, "vrn": { "email": "user@email.com", "installationId": 12345 }, "knx": { "host": "192.168.1.50", "groupAddresses": [ { "address": "1/1/1", "dpt": "DPT9", "label": "Living Temp" } ] }, "ffmpegRtsp": { "enabled": true, "basePort": 8554 }, "homekit": { "pin": "031-45-154" }, "server": { "port": 3001, "letsEncrypt": { "enabled": true, "domain": "lsh.yourdomain.com" } } }
```

TECHNICAL SPECIFICATIONS

PROPERTY	VALUE
Runtime	Node.js 18 LTS or newer
Hardware	Raspberry Pi 4 (1 GB+) / any Linux x64
RAM footprint	≈ 120 MB idle / < 250 MB under load
Disk	~50 MB app + ~10 MB HomeKit persist
Real-time	Socket.IO 4 push — sub-100 ms UI updates
TLS	Let's Encrypt ACME (auto-renew) or manual cert
HomeKit	HAP-nodejs — unlimited accessories
MQTT	MQTT.js v5 — reconnect, retained, QoS
Modbus	modbus-serial TCP — multiple devices
KNX	knx npm — DPT1/5/9/14 encode+decode
RTSP proxy	FFmpeg re-stream — on-demand, per-camera
Auth	bcrypt hashed users — setup wizard on first run
Config	config.json + Settings UI — no YAML
Persistence	HomeKit state in <code>persist/</code> — survives restarts

REST API — SELECTED ENDPOINTS

METHOD	PATH	DESCRIPTION
GET	/api/status	Full snapshot: connection source (MQTT/VRM), relay states, sensor values, platform list
GET	/api/sensors	All registered sensor entities with current values, units, HomeKit type, and controllable flag
POST	/api/relay/:index	Toggle relay 0 or 1. Body: <code>{"state": true false}</code> . Works over MQTT or VRM.

METHOD	PATH	DESCRIPTION
POST	/api/sensor/:key/command	Send a command to any controllable sensor (switch, light, thermostat). Body: <code>{"value": ...}</code>
GET	/api/rtsp-proxy	List FFmpeg RTSP proxy streams: name, port, slug, source URL, active status
GET	/api/cameras	UniFi Protect camera list with snapshot URLs and motion event feed
POST	/settings/mqtt	Save Victron MQTT broker settings (host, port, portallid)
POST	/settings/knx	Save KNX gateway config and group address list with DPT types
POST	/settings/smartbob	Save SmartBob MQTT broker and entity definitions (topic, type, label)
POST	/settings/test-*	TCP connectivity test for any integration before saving (returns latency ms or error)
POST	/settings/ffmpeg-rtsp	Enable/disable FFmpeg proxy, set base port and ffmpeg binary path
POST	/api/rooms	Save room definitions — name, icon, and list of sensor/device keys assigned to each room



Deployment Options

- Bare metal** Raspberry Pi 4 running Raspberry Pi OS Lite — recommended for energy monitoring close to Victron hardware
- Docker** Single container — mount `config.json` and `persist/` as volumes; expose ports 3001 + HomeKit HAP port
- NAS / VM** Synology DSM, Proxmox LXC, TrueNAS Jail — any Linux environment with Node.js and network access to MQTT broker
- systemd service** Auto-start on boot, journal logging, automatic restart on crash — one-command service install included



What Makes LSH Different

- **No Home Assistant dependency** — standalone Node.js, minimal attack surface
- **Victron-first** — designed for off-grid / solar installations, not just consumer IoT
- **MQTT bus explorer** built in — no separate MQTT.fx or MQTT Explorer needed
- **On-demand RTSP proxy** — no bandwidth wasted when no one is watching cameras
- **KNX bus native** — first-class KNXnet/IP support with per-address DPT config
- **Settings UI** — add devices through browser forms, not YAML or Python scripts
- **SIP softphone** — browser-based VoIP so you can answer the door intercom from the dashboard



LSH

Strategic Analysis — SWOT

BUSINESS ANALYSIS

Strengths · Weaknesses ·
Opportunities · Threats



Strengths

INTERNAL · POSITIVE

- ◆ **Local-first, zero cloud dependency**
All data stays on LAN. MQTT runs directly to hardware — no relay server, no account, works during internet outages.
- ◆ **20+ integrations in a single process**
~120 MB RAM footprint. No microservice sprawl, no Docker Compose with 12 containers — just `node server.js`.
- ◆ **Victron Energy depth**
Solar, battery SoC, grid, relay control — uniquely positioned for off-grid and solar installations. SolarEdge overlay included.
- ◆ **Native Apple HomeKit bridge**
Every sensor, switch, and camera auto-exposed. Unlimited accessories — survives restarts without re-pairing.
- ◆ **Protocol breadth**
MQTT, KNXnet/IP, Modbus TCP, REST, WebSocket, RTSP — standard protocols only, no proprietary SDKs.
- ◆ **Settings UI — no YAML required**
All integrations configured via browser forms with live connectivity tests. KNX group addresses, SmartBob topics, FFmpeg proxy — all point-and-click.



Weaknesses

INTERNAL · NEGATIVE

- ▲ **Technical setup required**
No one-click installer or Docker Hub image. Requires Node.js, terminal comfort, and manual `config.json` editing for initial setup.
- ▲ **No automation engine**
No built-in rule or scene builder. Users must rely on external tools (Node-RED, cron, HomeKit automations) for time or condition-based triggers.
- ▲ **No persistent history / charting**
Live values only — no time-series database, no energy graphs over days or weeks. Historical data requires InfluxDB + Grafana sidecar.
- ▲ **Single-node, no HA failover**
One server failure = no dashboard, no HomeKit bridge. No clustering or standby replication supported.
- ▲ **Cloud API fragility**
SmartThings, LG ThinQ, and VRM integrations depend on vendor API stability. Unilateral breaking changes can silently disable features.
- ▲ **Single maintainer bus factor**
Platform continuity depends on one developer. No community governance, no contributor pipeline, no public issue tracker yet.



Opportunities

EXTERNAL · POSITIVE

- ◆ **Privacy-conscious market growth**
Users increasingly reject cloud-dependent hubs after outages (SmartThings 2022, Google Nest 2023). LSH's local-first story is a direct answer.
- ◆ **Victron / solar / EV adoption rising**
Victron Energy is the dominant brand in off-grid solar. EV charger integration + battery relay control gives LSH a unique energy management angle no competitor covers.
- ◆ **KNX building automation market**
KNX is the dominant bus in European commercial buildings. First-class KNXnet/IP support opens the professional integrator and architect market.
- ◆ **Docker image + NAS packaging**
A published Docker Hub image and Synology/Unraid community package would unlock a large segment of homelab users who deploy from package managers.
- ◆ **Commercial integrator channel**
KNX + Victron + Loxone overlap with professional AV and building automation installers — a channel underserved by Home Assistant's DIY-first positioning.



Threats

EXTERNAL · NEGATIVE

- **Home Assistant dominance**
Enormous ecosystem (3,000+ integrations), massive community, strong brand, and well-funded Nabu Casa behind it. Mindshare is very hard to displace.
- **Vendor API changes**
SmartThings, LG ThinQ, and VRM can revoke or break APIs without notice. No contractual SLA between LSH and any vendor platform.
- **Matter / Thread commoditisation**
If Matter becomes universal, Apple Home and Google Home absorb device control natively — reducing the differentiation of multi-protocol middleware like LSH.
- **Security exposure at network edge**
Self-hosted dashboards port-forwarded to the internet are a common attack vector. Misconfigured TLS or weak passwords can expose the entire home network.
- **Competing open platforms**
openHAB, Domoticz, ioBroker, and Gladys cover overlapping protocol breadth with established communities and package registries.

◆ Automation engine add-on

Adding a Node-RED-style visual flow builder or a simple rule engine (if sensor X > value → command Y) would significantly expand the non-technical user segment.

● HAP-nodejs / HomeKit spec changes

The HAP bridge relies on a reverse-engineered Apple spec. Apple can introduce MFi restrictions or firmware changes that break the bridge without warning.